BeatAhead Phase 6 Model Freeze & Deployment Readiness Report

Document: `reports/phase6_deployment_readiness.md`
Execution Date: 2026-09-22 19:24:00
Model Name: `beatahead_phase5_model` (`v1.0.0`)
Status: FROZEN & PACKAGED — READY FOR STAGING INTEGRATION (WEBSITE CODE UNTOUCHED)

1. Executive Summary

Phase 6 Model Freeze and Deployment Packaging is complete. The exact deployment model, preprocessing pipeline, feature schema, decision threshold, and checksum manifests have been permanently serialized to `models/`. A production-grade inference engine (`src/inference.py`) and a comprehensive independent test suite (`tests/test_inference.py`) have been created and verified.

2. Selected Deployment Candidate Specification

DEPLOYMENT CANDIDATE SPECIFICATION

Specification Item	Exact Frozen Value
Model Name	beatahead_phase5_model
Version	1.0.0
Machine Learning Algorithm	XGBClassifier (Extreme Gradient Boosting Trees)
Framework & Version	xgboost v3.4.1 (Python 3.13)
Primary Artifact Path	models/beatahead_phase5_model.joblib
Artifact File Size & SHA-256	43,693 bytes | 528ff3f8f5edac6f3baf5aef8715d5e86f478462...
Feature Matrix	Matrix A (Full Multimodal + Pre-Event ST)
Feature Count	Exactly 26 features
Feature Schema Path	models/feature_schema.json
Metadata Manifest Path	models/MODEL_MANIFEST.json
Training Dataset Version	VitalDB 100-Case Cohort (80 Development Cases, 8,571 windows)
Preprocessing Pipeline	SimpleImputer(strategy='median') fit on 80 Dev cases
Deployment Decision Threshold	0.156742 (calibrated via Dev out-of-fold F1 maximization)
Alert Lead Time Guarantee	5-minute blank gap preceding 5-minute target evaluation

3. Frozen Feature Order (26 Features)

The inference pipeline strictly enforces the canonical 26-feature sequence defined in `models/feature_schema.json`:

```
 1. ecg_hr_mean          [bpm]     Mean heart rate over 5-minute observation window
 2. ecg_hr_std           [bpm]     Heart rate standard deviation (autonomic variability)
 3. ecg_rr_sdnn          [ms]      SDNN: Standard deviation of normal-to-normal RR intervals
 4. ecg_rr_rmssd         [ms]      RMSSD: Root mean square of successive RR interval differences
 5. ecg_pnn50            [%]       Percentage of adjacent RR intervals differing by > 50 ms
 6. ecg_r_amp_mv         [mV]      Median R-wave peak amplitude above isoelectric baseline
 7. ecg_qrs_width_ms     [ms]      Estimated QRS complex duration
 8. ecg_sqi              [ratio]   ECG signal quality index (fraction of clean beats)
 9. pat_median_ms        [ms]      Median Pulse Arrival Time (QRS peak to PPG foot delay)
10. pat_iqr_ms           [ms]      Interquartile range of PAT (vascular compliance stability)
11. pat_valid_fraction   [ratio]   Fraction of cardiac cycles yielding physiological PAT
12. pat_valid            [binary]  Binary flag indicating >= 30% valid PAT beats
13. ppg_pulse_amp        [a.u.]    Median photoplethysmogram peak-to-trough pulse amplitude
14. ppg_perfusion_index  [%]       Photoplethysmogram Perfusion Index (AC/DC ratio)
15. ppg_crest_time_ms    [ms]      PPG foot to systolic peak crest time (vascular stiffness)
16. ppg_sqi              [ratio]   PPG signal quality index (clipping and saturation detection)
17. spo2_mean            [%]       Mean pulse oximeter arterial oxygen saturation
18. spo2_min             [%]       Minimum arterial oxygen saturation
19. spo2_std             [%]       Standard deviation of pulse oximeter saturation trend
20. spo2_desat_count     [seconds] Duration of desaturation episodes (SpO2 <= 90%)
21. st_obs_mean          [mm]      Mean ST segment deviation in Lead II
22. st_obs_median        [mm]      Median ST segment deviation in Lead II
23. st_obs_min           [mm]      Minimum ST segment deviation in Lead II
24. st_obs_std           [mm]      Standard deviation of ST segment in Lead II
25. st_delta_baseline    [mm]      ST shift relative to initial 10-minute surgical baseline
26. st_slope_mm_min      [mm/min]  ST segment linear regression slope across observation window
```

4. Expected Input & Output Format

Input Format (`predict_single_window`):

Accepts a dictionary, pandas Series, or single-row DataFrame containing all 26 feature keys:

```
sample_input = {
    "ecg_hr_mean": 75.4,
    "ecg_hr_std": 6.8,
    "ecg_rr_sdnn": 55.2,
    "ecg_rr_rmssd": 42.1,
    "ecg_pnn50": 8.5,
    "ecg_r_amp_mv": 0.85,
    "ecg_qrs_width_ms": 95.0,
    "ecg_sqi": 0.98,
    "pat_median_ms": 220.0,
    "pat_iqr_ms": 45.0,
    "pat_valid_fraction": 0.85,
    "pat_valid": 1,
    "ppg_pulse_amp": 32.5,
    "ppg_perfusion_index": 85.0,
    "ppg_crest_time_ms": 110.0,
    "ppg_sqi": 1.0,
    "spo2_mean": 99.5,
    "spo2_min": 98.0,
    "spo2_std": 0.4,
    "spo2_desat_count": 0,
    "st_obs_mean": -0.45,
    "st_obs_median": -0.50,
    "st_obs_min": -0.85,
    "st_obs_std": 0.15,
    "st_delta_baseline": -0.35,
    "st_slope_mm_min": -0.02,
}
```

Output Format:

Returns a structured dictionary with continuous probability and binary decision thresholding:

```
{
    "model_probability": 0.284152,        # Continuous probability in [0.0, 1.0]
    "calibrated_probability": 0.284152,   # Probability calibrated for logloss
    "decision_threshold": 0.156742,       # Frozen Phase 5 deployment cutoff
    "binary_early_warning": 1,            # 1 = Impending Ischemia, 0 = Normal
    "alert_status": "ALERT",              # "ALERT" if prob >= threshold else "NORMAL"
    "lead_time_minutes": 5,               # Guaranteed advance lead time
    "model_version": "1.0.0",             # Model artifact release tag
    "feature_schema_version": "1.0.0",    # Feature schema version
    "model_name": "beatahead_phase5_model"
}
```

Strict Input Rejection Rules:

The engine (`src/inference.py`) strictly raises `FeatureValidationError` upon: - Missing any of the 26 required features. - Including extra unexpected features. - `NaN` or infinite (`inf`) values in any feature. - Non-numeric strings or `None` values. - Values violating defined physiological ranges (e.g., `ecg_hr_mean` $> 300\text{ bpm}$ or `spo2_mean` $> 100\%$).

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5. Verification & Test Suite Results

The independent test suite (`tests/test_inference.py`) was executed in a clean Python process:

```
Ran 11 tests in 1.592s

OK
```

- **Test 1 (Valid Input)**: PASS (All output fields present, probability in $[0, 1]$).
- **Test 2 (Missing Feature)**: PASS (`FeatureValidationError` raised).
- **Test 3 (Extra Feature)**: PASS (`FeatureValidationError` raised).
- **Test 4 (NaN Value)**: PASS (`FeatureValidationError` raised).
- **Test 5 (Infinity Value)**: PASS (`FeatureValidationError` raised).
- **Test 6 (Wrong Data Type)**: PASS (`FeatureValidationError` raised).
- **Test 7 (Deterministic Repeated Inference)**: PASS (10 runs yielded bit-for-bit identical probabilities).
- **Test 8 (Feature Order Invariance)**: PASS (Shuffling dictionary keys yielded identical probability).
- **Test 9 (Threshold Logic)**: PASS ($\ge 0.156742 \to \text{ALERT}$, $< 0.156742 \to \text{NORMAL}$).
- **Test 10 (Clean Process Loading)**: PASS (Joblib model and JSON schema load cleanly).
- **Test 11 (Physiological Bounds Check)**: PASS (Out-of-range HR and SpO2 strictly rejected).
- **Reproducibility Verification (TASK 8)**: Confirmed identical output across independent fresh Python process invocations.

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6. Known Limitations & Failure Modes

1. **Insidious vs. Abrupt Onset Sensitivity**: - The model is highly sensitive to insidious, gradual pre-event ST micro-sagging ($\text{mean ST} = -0.53\text{ mm}$ during T_{obs}). - In patients where ischemia occurs precipitously without pre-event drift ($\text{mean ST} = +0.27\text{ mm}$ during T_{obs}), the model remains silent to prevent false alarms, achieving **99.7% specificity**. A real-time threshold reflex ($\text{ST} \le -1.0\text{ mm}$) must operate alongside the model.
2. **Dependence on ST Electrophysiology**: - As proven by the Matrix B ablation benchmark (AUROC 0.53), non-ST features alone (PAT, HRV, pulse morphology) cannot reliably forecast intraoperative ischemia. ST segment Lead II inputs are required.
3. **Single Lead II Scope**: - The model monitors Lead II. Purely anterior/lateral events manifesting exclusively in precordial leads (e.g. V5) without Lead II involvement are outside the single-lead inferential scope.
4. **Motion Artifacts & Electrocautery**: - High-noise intervals failing SQI must be rejected prior to inference.

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7. Next Steps & Deployment Readiness

- The serialized model artifact (`models/beatahead_phase5_model.joblib`), schema (`models/feature_schema.json`), metadata (`models/model_metadata.json`), and manifest (`models/MODEL_MANIFEST.json`) are frozen and sealed.
- All code on the BeatAhead website remains 100% untouched.
- Staging and deployment integration will proceed only upon user authorization.